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Ensemble Learning Model for Deepfake Audio Detection using Multi-Feature Extraction Approach

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Abstract: In this paper, several audio features are leveraged to improve detection accuracy through an advanced ensemble learning approach. For the extraction of diverse signal characteristics, the method uses Mel-Frequency Cepstral Coefficients (MFCCs), statistical features, and Short-Time Fourier Transform (STFT) features. Specialized neural networks are utilized to handle each set of features: MLPs categorize speech according to statistical descriptions, RNNs detect temporal anomalies from MFCC, and CNNs examine spectral patterns from STFT. These are finally combined by a meta-learning model through the stacking ensemble method, enhancing robustness and decreasing misclassification. Compared to individual models, the system is more effective when evaluated against the Fake-or-Real dataset. Based on experimental findings, deepfake detection is significantly enhanced and highly accurate when spectral, temporal, and statistical data are fused. In addition, the model is hosted using Gradio on Hugging Face, providing real-time deepfake detection.

Keywords: Deepfake detection, ensemble learning, audio forensics, neural networks, STFT, MFCC, meta-learning

I. INTRODUCTION

Artificial intelligence (AI) has changed many industries, from healthcare to entertainment, bringing both great benefits and serious ethical concerns. One of the most talked-about advancements in AI is deepfakes—computer-generated media that can realistically copy a person’s voice, facial expressions, and movements, making them appear to say or do things they never actually did. Deepfake videos have received a lot of attention, but the emergence of audio deepfakes is just as concerning. These artificial intelligence (AI)-generated voice recordings remarkably accurately mimic human speech, including speech patterns, pitch, and tone. It is practically hard to tell the difference between real and false audio because AI models can create completely new sentences in the speaker's voice from just a few minutes of recorded speech [1].

Despite the risks, voice cloning technology has many uses. AI-generated speech can help those with speech issues

to communicate more naturally by restoring their own voices [2]. It is possible to make digital assistants like Alexa and Siri sound more individualized and human [3]. Deepfake audio has the potential to increase production efficiency in the film and media industries by removing the need for actors to re-record dialogue [4].

But there are also severe dangers from the same technology that enables these advances. In a highly publicized 2019 case, hackers duped a company into sending

€220,000 to a fake account by using AI-generated speech impersonation [5]. This case demonstrates the increasing threat posed by deepfake audio in financial fraud. Political misinformation is also a significant issue, as manipulated audio clips of global leaders making false statements will be shared on social media, sway public opinion, and even cause anarchy [6]. Deceptive speech recordings are employed to damage reputations or wrongly accuse individuals of crimes, and deepfake audio is also employed for extortion and slander [7].

It can be quite challenging to identify audio deepfakes because many AI-generated voices are nearly indistinguishable from the real ones. Since manual detection is usually not effective, advanced AI-based detection tools are required [8].

Feature extraction is the first step towards detecting deepfake audio, where significant audio signal features are analyzed for anomalies. Popular techniques are Short-Time Fourier Transform (STFT), which decomposes audio signals into time-divided frequency information [1]; Mel-Frequency Cepstral Coefficients (MFCC), popularly employed in speech recognition [4]; and Chroma Features and Zero-Crossing Rate (ZCR), used to identify unnatural frequency changes [3].

After the extraction of these characteristics, AI algorithms trained to discriminate between artificially synthesized and natural voices are employed for analysis. Convolutional Neural Networks (CNNs) identify spatial patterns within

spectrograms [7], while Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) models note temporal correlations among speech patterns [6]. Multi-Layer Perceptrons (MLP) also assist in audio sample classification by using recovered features [8].

To improve accuracy, modern detection systems integrate several AI models. These systems acquire a thorough grasp of the audio properties by combining various feature extraction methods. After that, the classification layer establishes if an audio sample is real or AI-generated. This multi-layered strategy ensures that the system continues to be effective against developing deepfake technologies by increasing detection accuracy, scalability, and adaptability [2]. Incorporating AI-driven detection models improves the detection of deepfake audio while reducing possible hazards and shielding people, companies, and governments from harmful use of this technology.

The contribution of our work is based on the following:

- **A systematically engineered multi-representation feature extraction framework combining STFT, MFCCs, and statistical descriptors, designed to capture complementary spectral, temporal, and global signal characteristics for improved deepfake audio detection within the evaluated benchmark dataset.**
- A multi-model deep learning system using Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs) with LSTM layers, and Multi-Layer Perceptrons (MLPs), each being optimized for a different feature representation—CNNs for recognizing spectral patterns, RNNs for modeling temporal sequences, and MLPs for classification based on statistical features.
- Automated hyperparameter tuning using Keras Tuner's Hyperband algorithm to optimize the architectures for filter sizes, LSTM units, dense layer sizes, and learning rates for best classification accuracy and model generalization.
- A stacking ensemble learning strategy, with the CNN, RNN, and MLP model outputs aggregated into a fully connected meta-classifier based on the complementary strengths for more profound deepfake speech detection.

- In addition to system deployment, this work emphasizes structured empirical evaluation of feature-domain interactions within a stacking-based ensemble framework.

The rest of the paper is structured as mentioned further. Section 2 is a literature review of existing deepfake detection methods and their limitations. Section 3 is a description of the methodology as well as our feature extraction and AI-based classification processes. Section 4 is a discussion of

performance analysis and experimental results. Section 5, the conclusion of the study, declares potential directions for deepfake detecting technologies to evolve.

II. LITERATURE REVIEW

Growing complexity in synthetic speech generated by text-to-speech (TTS), voice conversion (VC), and generative adversarial networks (GANs) is a primary challenge for voice verification and content authentication systems. The response has come in the form of audio deepfake detection methods that use sophisticated feature extraction, resilient model architectures, and ensemble-based learning approaches. Hamza et al. [9] discussed several approaches to detecting synthetic speech, highlighting the necessity for better detection models. This survey consolidates major breakthroughs in the research area, making use of newly published works, which discuss some of the key modeling methods, datasets, and testing schemes. Feature engineering is still one of the significant components of audio deepfake detection.

The earliest works made extensive use of traditional features like hand-crafted Mel-Frequency Cepstral Coefficients (MFCCs), emulating human auditory senses and incorporating timbral information within speech. Hamza et al. [9], Priya et al. [10], and Kapoor et al. [11] employed these features with SVM, Random Forest, and Decision Trees classifiers. In Hamza et al. [9]'s research, MFCCs were extracted from the Fake-or-Real (FoR) dataset and tested on four subsets. Classic models such as SVM worked best with short-duration video clips, while deep models like VGG-16 were superior to others on the entire dataset, proving the strength of incorporating MFCCs with convolutional networks.

But MFCCs are not sufficient for identifying more subtle synthetic sounds. Recent works have combined features to obtain more comprehensive representations. Krishnan and Krishnan's [12] MFAAN model combined MFCCs, Linear-Frequency Cepstral Coefficients (LFCCs), and Chroma-STFT features with a multi-path CNN architecture. This architecture permitted the model to process spectral, harmonic, and temporal parts in isolation before combination, with higher accuracy on In-the-Wild and 94.47% on Fake-or-Real.

Pham et al. [13] advanced this method by obtaining multiple spectrograms with STFT, Constant-Q Transform (CQT), and Wavelet Transforms, and applying auditory filters such as Mel, Gammatone, and DCT. These spectrograms were taken as inputs to end-to-end models, transfer learning architectures (ResNet-18, MobileNetV3), and audio-pretrained models such as Whisper and Speechbrain. Their ensemble of multi-pipelines scored an Equal Error Rate (EER) of 0.03 on the ASVspoof 2019 dataset, ranking among the top-performing systems.

Yang et al. [14] pushed the line further by proposing a multi-view detection system. Their framework integrated handcrafted features (MFCCs, LFCCs, CQT) with pre-trained model embeddings (Wav2Vec2, HuBERT, Whisper, WavLM). A Transformer-based attention mechanism dynamically chooses and combines features, allowing the

model to be flexible to various kinds of audio manipulation. Tests on ASVspoof 2019, ASVspoof 2021, and the In-the-Wild dataset proved that learning-based features achieved improved generalization, especially under out-of-domain conditions.

Other research tested feature learning in harder scenarios. Rahman et al. [15] used Linear Filter Banks (LFBs) with higher high-frequency resolution compared to

Mel filters. They also added speech activity detection (SAD) for the elimination of silent regions that would otherwise bias the learning. The modified ResNet (x-ResNet)-based model, along with Probabilistic Linear Discriminant Analysis (PLDA) scoring, reported a 25% error reduction in classification on partially synthesized datasets with robust generalization across noisy and manipulated inputs.

Model selection has a central bearing on detection efficiency and accuracy. Standard machine learning models can be effective in particular conditions. A feature-engineered pipeline based on six classifiers with training from the Fake-or-Real dataset was employed by Iqbal et al. [16]. It produced its top-performing system with 26% gain over baseline systems for detection accuracy. Kapoor et al. [11] and Priya et al. [10] also used Random Forest classifiers over MFCC features, confirming the strength of reduced models when used with effective preprocessing and cross-validation.

Ensemble learning has become a strong method for performance enhancement and prevention of overfitting. Thirumalai et al. [17] compared bagging (Decision Trees), boosting (AdaBoost), and stacking (CNNs, LSTMs, GBMs with a logistic regression meta-model). Their stacked ensemble had the best F1 scores and generalizability over a proprietary dataset of real and synthetic audio examples. Pham et al. [12] also performed model-level ensembling through mean-fusion to perform ensemble-level fusion of deep architecture predictions and audio embeddings.

Deep learning remains supreme in high-performing environments. Almutairi and Elgibreen [1] presented Arabic-AD, a self-supervised Modern Standard Arabic (MSA) model that was trained on a small synthetic dataset. By employing wav2vec-style representation learning, the model reported 97% accuracy and an EER of 0.027%, demonstrating how self-supervision can achieve high performance even under data constraints. Krishnan and Krishnan's MFAAN [12] and Yang et al.'s multi-view Transformer [14] both employed deep feature extractors and late fusion networks to obtain competitive performance on several datasets.

Transfer learning is now also extensively used to borrow the prowess of vision or speech recognition models. Pham et al. [13] was also able to transform ResNet-18 and MobileNetV3, originally proposed for image-based tasks, for use with audio classification with training from the input space spectrogram. Meanwhile, other models such as Whisper, HuBERT, and Wav2Vec2 are currently applied in deriving embeddings directly embeddable by feeding to slim MLPs or ensemble classifiers [12][13].

Strong dataset construction and benchmarking are critical to enable meaningful assessment. The Fake-or-Real (FoR) dataset [9][16] is still popular because it has a range of

synthetic audio types and subsets (e.g., for-2sec, for-norm, for-rerec). ASVspoof 2019 and 2021 [13][14] are widely used benchmarks for assessing spoofing attacks on speaker verification systems. Pham et al. [13] and Yang et al. [14] both tested their models on these datasets, enhancing their real-world relevance.

A few studies introduced bespoke datasets. Almutairi and Elgibreen [18] designed a Modern Standard Arabic corpus to address the gap in multilingual deepfake detection. Asha et al. [19] built a multimodal dataset with visual and audio manipulation to train their D-Fence framework that employs bi-directional recurrent convolutional networks and cross-modal attention. Although dedicated to multimodal detection, the architecture provides insight into future cross-modal fusion approaches to audio.

Although datasets such as In-the-Wild [12][14] are realistic with their capture of uncontrolled situations, their variance is problematic to models trained for limited domains. Certain works, including Masood et al. [20], have highlighted how most systems cannot generalize across data sets, particularly when subjected to new synthesis scenarios or speakers. To solve this, Rahman et al. [15] and Yang et al. [13] included the use of data augmentation methods, including compression,

reverberation, and background noise, which made generalizability better.

A few models indicated high accuracy and low EERs. MFAAN [12] reported 98.93% accuracy on the In-the-Wild dataset and 94.47% on Fake-or-Real. Arabic-AD [18], even with limited data, attained 97% accuracy and an EER of 0.027%. Pham et al.'s ensemble system [12] reported an EER of 0.03 on ASVspoof 2019, and Yang et al.'s multi-view Transformer [13] outperformed generalization across ASVspoof 2019, 2021, and In-the-Wild.

Classic models also performed well. Hamza et al.'s VGG-16 model [9] was best among traditional ML pipelines on the for-original subset. Detection systems based on SVM, as indicated in several studies [9][19], reported up to 92% accuracy, 97% precision, and 88% recall, particularly when they were trained on short, stable audio samples.

Yet, not all studies presented total metric breakdowns. For instance, some studies [20][17][21] reported only accuracy or qualitative improvements without scores. This non-standardization makes direct comparison difficult and emphasizes the necessity for standardized benchmarking protocols.

Though promising, deepfake audio detection suffers from some chronic issues. The biggest of these is cross-dataset generalization. Models tend to excel on the datasets that they are trained upon but then deteriorate on new distributions. This is particularly a concern in cases of multilingual, low-resource, or novel synthetic models. Attempts by Yang et al. [14], Masood et al. [20], and Rahman et al. [15] emphasize the necessity for diverse, representative training sets as well as sophisticated generalization approaches.

Partial synthetic speech manipulation detection, where partial speech is replaced with synthetic speech, is the other important constraint. The models are mostly learned for fully

synthesized samples and don't perform as well in locating short insertions or hybrid attacks. Rahman et al.'s method [15] based on SAD and PLDA is one of the few efforts done to address the same, having a considerable decrease in error rate.

Real-time applicability is also left unexploited. Although models are highly accurate, they end up demanding excessive computational power. Pruning, audio-specific optimizations, and light-weight architectures are areas for potential future work [15][11][10].

III. METHODOLOGY

The proposed methodology for audio deepfake detection utilises an ensemble learning approach that integrates Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Multi-Layer Perceptrons (MLPs). The process of preprocessing of audio deepfakes entails obtaining useful features in speech recordings that can help allow deep learning architectures to identify deepfakes as well as the original audio. The process of feature extraction remains organised in an orderly manner, utilising time-frequency representations and statistical descriptors to take note of audio attributes.

A. Dataset Description:

The proposed method uses the benchmark Fake-or-Real (FoR) dataset that has more than 1,95,000 machine-generated speech and genuine human-made utterances **comprising both machine-generated speech and genuine human speech samples**. Classifiers could be trained to recognize fake speech on the basis of the given dataset.

There are four configurations of the given dataset available for use: for-original, for-norm, for-2sec, and for-rerec. **The for-original version contains the collected audio files in their raw format. The for-norm version includes normalization of sample rate, volume, and channel configuration, with balanced gender and class distribution. The "for-2sec" and "for-rerec" variants are 2-second truncated audio**

recordings derived from the normalized dataset, and a re-recorded variant of the "for-2sec" variant that simulates a scenario in which an attacker transmits an utterance over a voice channel (e.g., a voice message or phone call), respectively.

In this study, the dataset was partitioned using stratified sampling into training (64%), validation (16%), and testing (20%) sets, ensuring balanced class representation across splits. These two versions are applied to the intended investigation in order to accurately spot fine disturbances and manipulations in audio signals, which can be used by an attacker.

B. Data Preprocessing and Feature Extraction:

The system starts by loading raw audio data (.wav files). To transform this data into a suitable format for deep learning models, various audio features are extracted:

- **Short-Time Fourier Transform (STFT):** The computational process known as STFT converts a time-domain signal into a frequency-domain two-dimensional representation. It does so by preserving both time and frequency information through the application of the Fourier Transform to short, overlapping pieces of a signal. Convolutional Neural Networks (CNNs), which can use the structured 2D representation for pattern recognition, are greatly aided by their capacity to preserve both spectral and temporal fluctuations.
- **Mel-Frequency Cepstral Coefficients (MFCCs):** The spectral envelope of an audio stream, which is deeply related to how people perceive speech, is represented by MFCCs. These features are well-suited for detecting anomalies in false speech because they are often used in speech and speaker recognition. RNNs, particularly Long Short-Term Memory (LSTM) networks, are tasked with detecting time-series data's sequential relationships. The RNN can learn temporal patterns and anomalies in simulated speech since MFCCs retain phonetic and speech structure information.
- **1D Statistical Feature Vectors:** We extract six statistical features that summarize the global characteristics of the audio signal in addition to spectral and cepstral features. The rate at which the waveform crosses the zero-amplitude level is quantified by the Zero-Crossing Rate (ZCR), which provides details regarding speech texture. Root Mean Square (RMS) Energy captures the global loudness and amplitude variations of the signal. The measured pitch of sound is related to the center of mass, or spectral centroid, of the spectrum. Through measurement of how frequencies cluster about the spectral centroid, spectral bandwidth indicates the signal's richness in timbre. Spectral Rolloff indicates high-frequency emphasis by locating the frequency point below which lies 85% of the energy in the entire spectrum. Lastly, Spectral Flatness quantifies how noise-like or tonal a signal is by comparing the geometric and arithmetic means of the spectrum.

C. Model Training and Hypertraining Tuning:

To optimize model performance in detecting deepfake audio, we employ hyperparameter tuning using Keras Tuner's Hyperband algorithm. This process helps in identifying the best-performing configurations for three deep learning models: CNN, RNN, and MLP, each tailored to distinct feature representations.

The Convolutional Neural Network (CNN) is trained over Short-Time Fourier Transform (STFT) features to take advantage of its power to identify spatial patterns in spectrogram-like images. STFT gives a time-frequency domain description of the audio signal and thus is well adapted to CNNs, which are good at capturing localized

structures in 2D representations. For deepfake detection, CNNs are especially useful for detecting slight frequency changes and artifacts added during the process of synthetic audio creation. CNNs can learn discriminative features to differentiate between real and fraudulent audio thanks to these artifacts, which are typically undetectable in the time domain but manifest as distinct patterns in the spectrogram.

The two convolutional layers in the architecture are followed by MaxPooling for dimensionality reduction and feature extraction. While the kernel size is chosen from {3,5}, the number of filters in the convolutional layers is optimized within the range [16,128]. A fully connected layer with configurable units (64 to 256) is added before the sigmoid output layer after flattening. Additionally, the optimizer's learning rate is tuned from {1e-2, 1e-3, 1e-4}.

Mel-Frequency Cepstral Coefficients (MFCCs) are used to train the Recurrent Neural Network (RNN), which teaches it that speech signals are consecutive. RNNs, more especially Long Short-Term Memory (LSTM) or Gated Recurrent Unit (GRU) architectures, are very good at analyzing temporal dependencies and dynamic audio changes because speech is a time-series data type by nature. The spectral envelope of speech, which contains prosodic and phonetic information crucial for identifying abnormal speech patterns, is well-represented by MFCCs. RNNs can effectively detect deepfake audio due to their tendency to have unnatural transitions, inconsistent prosody, or lack of long-term coherence. We use an RNN with LSTM layers that is adjusted for the 40×80 MFCC feature matrices in order to capture the temporal dependencies in speech. The first LSTM layer returns sequences to enable hierarchical feature learning, and the second LSTM layer produces a last feature representation. The number of LSTM units in each LSTM layer is optimised in [32, 128]. Moreover, there is an added dense layer with adjustable units (64 to 256) prior to the sigmoid classifier.

Six 1D statistical feature vectors are used to train the Multi-Layer Perceptron (MLP) model. The recovered coefficients are then converted into a fixed-size numerical vector for classification. MLPs work with tabular numerical data and are best suited for learning intricate decision boundaries between real and fake audio, in contrast to CNNs and RNNs, which use spatial and temporal dependencies, respectively. Six statistical features—zero-crossing rate, root mean square energy, spectral centroid, bandwidth, roll-off, and flatness—are retrieved using the suggested method. An MLP model uses these condensed properties as inputs. The model picks up on spectral distributions and statistical patterns in the speech signal that point to deepfake characteristics. MLPs' fully connected structures enable them to detect nonlinear correlations between different spectral qualities and handle high-dimensional input information well. The final sigmoid output layer comes after two completely connected layers with configurable units ranging from [16, 128]. Because MLPs prioritize feature-based discrimination over sequential or spatial analysis, they provide yet another perspective on deepfake audio detection, which makes them especially helpful as complementary models in the ensemble technique.

The Hyperband algorithm in Keras Tuner is used for tuning every model. It chooses the best hyperparameter configurations adaptively based on validation accuracy. In order to limit effective exploration, not more than 10 epochs are utilized in a trial. Models that perform best in each tuning session are selected for the final evaluation. These three architectures—CNNs for spectral pattern identification, RNNs for temporal sequence modeling, and MLPs for statistical feature learning—are blended to form a diverse

ensemble, each with specialties. Their combined outputs are then aggregated using a meta-classifier to enhance detection robustness and overall performance in detecting deepfake speech.

For reproducibility, all models were implemented using TensorFlow/Keras and developed in the Google Colab environment. Model optimization was performed using the Adam optimizer with binary cross-entropy loss, while hyperparameter tuning was

conducted using Keras Tuner's Hyperband algorithm. During tuning, candidate models were evaluated using validation accuracy, and the best-performing configurations were selected for final evaluation. The final fusion model was trained for 20 epochs with a batch size of 32 using the training and validation partitions described in Section 3-A.

D. Prediction Stacking and Ensemble Learning

The ensemble framework is motivated by the complementary nature of the selected feature representations. STFT features capture spectral and time-frequency characteristics, MFCC features preserve temporal and phonetic information, and statistical descriptors summarize global signal behaviour. Since each feature domain provides different information about the audio signal, no single representation is expected to capture all artifacts associated with synthetic speech. The stacking strategy is therefore employed to combine these complementary sources of information and improve overall detection robustness.

The overall robustness and accuracy of deepfake audio detection are increased by using a stacking ensemble method to combine the outputs from the CNN, RNN, and MLP models. This technology creates a new feature vector by combining the discrete outputs of the three basic models: CNN, which recognizes spatial frequency patterns; RNN, which learns sequential dependence; and MLP, which is concerned with the statistical association of features. CNN and RNN features are flattened before concatenation to create a single, high-dimensional feature vector that normalizes input dimensions. Using stratified sampling, the dataset is further separated into groups for training (64%), validation (16%), and testing (20%). Every base model contributes its learned representations of the audio signal to this feature vector, which is a higher-level abstraction of the decision-making process. Rather than relying on any single model, stacking utilizes the complementary strengths of these architectures, mitigating the weaknesses of individual models and improving generalization across diverse deepfake variations.

A specific deep-learning model is trained using each feature set. Following spatial frequency pattern extraction of spectrograms through a two-layer convolutional network, the CNN model classifies through fully connected layers. Temporal dependencies in speech are learned through the RNN model by the convolution of MFCC time-series through a stacked Long Short-Term Memory (LSTM) network. Statistical features are learned by the MLP model, which is a light, fully connected network. Hyperparameter tuning is performed with Keras Tuner's Hyperband in order to optimize model performance through identifying the best filter sizes (CNN), LSTM units (RNN), dense layer sizes, and learning rates. Based on validation accuracy, the best-performing models are selected.

We use a feature-level fusion approach, in which the concatenated feature set is fed into a four-layer fully connected neural network, to take advantage of the strengths of all three architectures. The Adam optimizer with a learning rate of 0.001 is used to optimize this network after it has been trained using binary cross-entropy loss. The last layer uses a sigmoid activation function for binary classification and outputs a probability score.

E. Model Evaluation

Common performance metrics, such as accuracy, which measures the overall accuracy of prediction, precision and recall, which measure the ability to accurately detect forged audio, and the F1-score, which balances precision and recall for better evaluation, are utilized to assess the final ensemble model. The architecture is holistic in its approach to deepfake detection through the integration of multiple deep learning models, such as CNNs for spatial feature extraction, RNNs for sequential pattern analysis, and MLPs for statistical feature-based classification. To evaluate performance gains through feature fusion, the ensemble predictions of the fusion model are compared with the predictions of the CNN, RNN, and MLP

models. The advantages of utilizing multiple feature representations in deepfake detection are supported by the outcomes, which indicate that ensemble learning significantly enhances total classification performance.

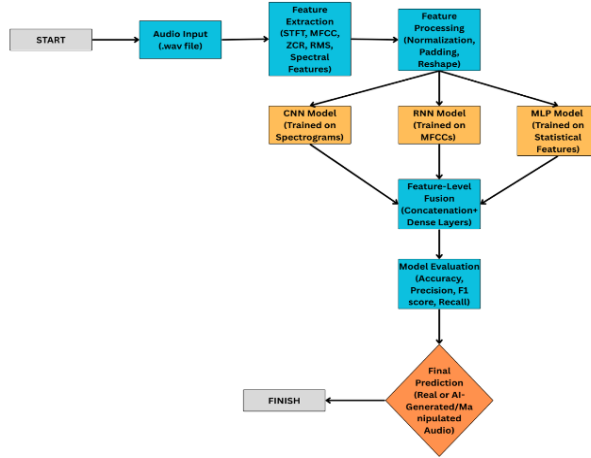


Figure 1. Flowchart of the proposed model

Figure 1 depicts a process for identifying authentic or artificial intelligence-generated/manipulated audio. It starts with a '.wav' file as the audio input, and then it extracts features (such as STFT, MFCC, ZCR, RMS, and spectral features) and processes them (like normalizing, padding, and reshaping). Then, three machine learning models are used: MLP (trained on statistical characteristics), RNN (trained on MFCCs), and CNN (trained on spectrograms). Concatenation and thick layers are utilized to merge their outputs at the feature level. Before producing a final forecast indicating whether the audio is real or artificial intelligence (AI)-generated/manipulated, the model is assessed based on accuracy, precision, F1 score, and recall.

F. Model Deployment

TensorFlow's model.save() is employed to save the trained CNN, RNN, and MLP deep learning models, while joblib.dump() is employed to save the ensemble model. These models are deployed in Hugging Face Spaces to ensure scalable and seamless access, allowing users to interact with the system whenever they want. Real-time inference is handled in Google Colab with Gradio during development

and testing, providing a lightweight and interactive environment before final deployment.

Three features are derived by the system that are critical to deepfake audio classification: six statistical features for MLP-based classification, mel-frequency cepstral coefficients (MFCCs) for RNN in order to recognize speech patterns, and log-magnitude short-time Fourier transform (STFT) for CNN to capture spectral information. Features retrieved are correctly formatted prior to being input to the models through a preprocessing stage. The CNN, RNN, and MLP all generate independent predictions that are concatenated into a feature vector as part of an end-to-end structured inference pipeline utilized in the classification task. The ensemble model employs this vector as input to make the classification decision by leveraging the complementary strengths of each model to enhance accuracy.

Gradio provides an easy-to-use web-based interface for users to interact with, enabling them to input audio files to be detected as deepfakes. The input is processed automatically, relevant features are extracted, inference is done with the trained models, and a classification label and confidence score are given back. Without needing dedicated hosting infrastructure, Google Colab generates a shared link that brings content within reach of the masses.

There are two stages to the deployment plan: a larger deployment on Hugging Face

Spaces for scalable and long-lasting access, and a temporary deployment on Google Colab for quick testing and live inference. The flexibility of cloud-based inference and the reliability of hosting over the long term are integrated in this hybrid solution. The solution delivers reliable real-time deepfake detection by integrating deep learning models with an intuitive interface, ensuring both ease of access and high accuracy.

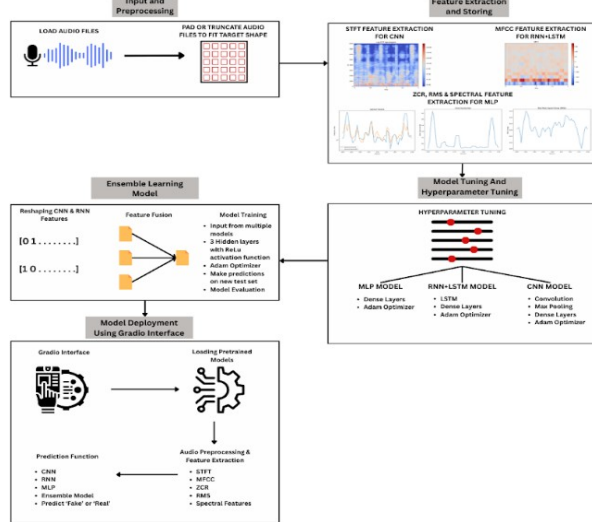


Figure 2. Architecture diagram of the proposed system

The above figure (**Figure 2**) consists of the architecture diagram for the entire proposed system. Audio files are first loaded, and to ensure uniformity, they are padded or trimmed into a certain shape. Feature extraction is the second step, where CNN employs STFT features, RNN+LSTM employs MFCC features, and MLP employs ZCR, RMS, and spectral features. All models are both model and hyperparameter adjusted. With thick layers and the Adam optimizer for training and evaluation, the ensemble learning model integrates CNN and RNN characteristics. Finally, a Gradio interface is employed to deploy the model, loading pre-trained

models and identifying whether an audio file is "Fake" or "Real."

IV. RESULTS & DISCUSSION

The efficiency of combining Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Multi-Layer Perceptrons (MLPs) in a stacked ensemble architecture was proved by the better classifying performance of the proposed ensemble learning approach for detecting deepfake audio. To measure the precision of the system in distinguishing real from fake audio signals, evaluation measures like accuracy, precision, recall, and F1-score were computed. With an accuracy of 97.85% and balanced recall (95.98%) and precision (98.91%) values, these are the results of a highly successful deepfake detection system that correctly labels real and fake audio instances with minimal false positives and false negatives. The model's strength is also attested to by the F1-score (97.42%), which provides a harmonic mean of recall and precision.

To determine how each base model contributed to the overall detection performance, each one in the ensemble was assessed separately. With an accuracy of 93.92%, the CNN Model showed excellent performance in utilizing STFT characteristics to capture spectral patterns. The LSTM-based RNN Model demonstrated its efficacy in learning sequential dependencies from MFCC features with an accuracy of 85.86%. With an accuracy of 82.8%, the MLP Model demonstrates that while statistical features by themselves can offer useful indicators for deepfake identification, they are not as successful as spectrogram-based methods.

The independent evaluation of CNN (STFT), RNN (MFCC), and MLP

(statistical features) provides an indicative comparison of feature-domain contribution. While this does not constitute a strict ablation study with systematic feature removal, the performance differences suggest complementary strengths across spectral, temporal, and statistical representations.

The ensemble model achieved higher accuracy than the individual models, suggesting that combining complementary feature representations enhances detection performance. The improved performance of the stacked model can be attributed to the complementary strengths of CNN-based spectral analysis, RNN-based temporal modeling, and MLP-based statistical feature learning. This suggests that errors made by individual models are partially complementary, allowing the ensemble framework to exploit information that may not be fully captured by any single feature representation. The proposed technique demonstrated strong generality for multiple deepfake audio types, including voice conversion, synthesized speech, and modified recordings. Through reduced overfitting and improving the ability of the model to generalize to new, unseen data, the stacking ensemble technique helped reduce the potential shortfalls of each model individually.

Although the evaluation dataset includes the re-recorded ("for-rerec") subset that partially reflects real-world transmission effects, the present study does not provide a dedicated robustness analysis under controlled noise conditions, compression artifacts, or partially manipulated speech segments. Future work will investigate the performance of the proposed framework under these challenging real-world scenarios to further assess its generalization capability.

For real-world deployment, the model was successfully integrated into an interactive interface using Gradio on Hugging Face, allowing users to upload and analyze audio samples in real-time.

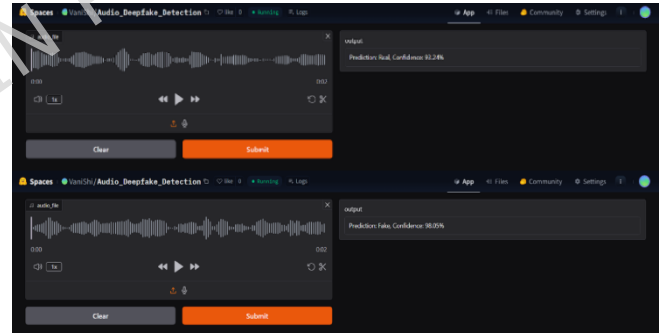


Figure 3. Web-based interface for model deployment

Figure 3 displays an audio deepfake detecting application's Gradio-based user interface. In this interface, audio files can be recorded or uploaded by the user and displayed as a waveform, as shown. Upon selecting "Submit," the audio is processed by the system, which then displays a forecast in the output box. The audio is categorized as "Real" or "Fake" with a calculated confidence score, according to the findings. Buttons for playback, input clearing, and uploading a new file are all included in the interface.

The following table shows a comprehensive comparison of the proposed work with existing models. The comparison is primarily based on accuracy values to maintain uniformity in cross-study evaluation:

TABLE I. COMPARISON OF RESULTS OBTAINED

IT IS IMPORTANT TO NOTE THAT DIRECT NUMERICAL COMPARISON ACROSS STUDIES SHOULD BE INTERPRETED CAUTIOUSLY, AS DIFFERENT WORKS EMPLOY VARYING DATASETS, PREPROCESSING PIPELINES, AND EVALUATION SPLITS. THE COMPARISON PRESENTED IN TABLE 1 IS THEREFORE INDICATIVE RATHER THAN STRICTLY BENCHMARK-EQUIVALENT. THE PRIMARY PURPOSE IS TO CONTEXTUALIZE THE PERFORMANCE RANGE OF RECENT DEEPFAKE DETECTION SYSTEMS RATHER THAN CLAIM DIRECT SUPERIORITY UNDER IDENTICAL

N	Pap	Auth	Y	Datas	Model	Accu
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EXPERIMENTAL CONDITIONS.

	with a Multi-Feature Authenticity Network	an, K. S.				
4	Deepfake Audio Detection Using Spectrogram-based Feature and Ensemble of Deep Learning Models	Pham, L., et al	2024	ImageNet1K	CNN, ConvNet-Tiny, Whisper	90
5	Deepfake Audio Detection via Feature Engineering and Machine Learning	Iqbal, F., et al.	2022	Fake-or-Real	XGB	93
6	Deepfake Detection Using Ensemble Model Techniques for Audio Analysis	Thirumalai G., Kadir A., Malaram M. Naidu, Thariq A. Khan, Venkatesan M.	2024	Kaggle, UCI Machine Learning Repository	Bagging, Boosting (AdaBoost), Stacking (Logistic Regression)	97% (Bagging), 94% (Boosting), 95.67% (Stacking)
7	Detecting Fake Audio of Arabic Speakers Using Self-Supervised Deep	Zaynab M. Almutairi, Hebah Elgibre	2025	Ar-DAD, Arabic Speech Corpus (ASC), Arabic-CAPT	Arabic-AD (Self-Supervised Learning with HuBERT)	97% (All Tasks)

indicated that, compared to individual architectures, combining numerous deep learning models significantly improves classification performance within the scope of the Fake-or-Real dataset.

The test proves that CNNs work well for spectral pattern analysis, RNNs perform best in temporal inconsistencies, and MLPs yield complementary statistical information. The stacking ensemble approach effectively integrates complementary feature domains to achieve improved classification performance within the evaluated dataset.

In spite of high accuracy, the model also has some limitations. The proposed work is limited by the scope of the dataset and needs to be further validated using cross-dataset evaluation and diverse statistical measures. While a real-time deployment prototype has been implemented, extensive validation under diverse real-world deployment environments remains part of future work. The ensemble method needs several deep learning models, which raises inference time over using single-model designs. The model might be vulnerable to adversarial attacks, where minor manipulations in synthetic speech can mislead detection systems. Although the model worked well on high-quality audio, performance when the audio is noisy or low-bitrate should be examined further. Future research will aim to enhance efficiency by speeding up model inference, increasing robustness against manipulations by adversaries, and expanding testing to real-

world data with different degrees of noise and distortion.

In summary, this comprehensive study presents a systematically engineered ensemble framework for deepfake audio detection, demonstrating strong detection performance within the evaluated benchmark and demonstrating practical feasibility through prototype deployment.

This framework not only enhances the authenticity of audio content but also significantly bolsters the security of speech-based communication systems, ensuring that users can trust the integrity of their communications in an increasingly digital and interconnected world.

While a prototype deployment was successfully implemented using Gradio and Hugging Face Spaces, detailed analysis of inference latency, computational resource consumption, and large-scale deployment scalability was not conducted in the present study and remains part of future work.

DATA AVAILABILITY STATEMENT

The datasets generated and/or analysed during the current study are available in the Fake-or-Real Dataset. on Kaggle. Link to the dataset used:

<https://www.kaggle.com/datasets/mohammedabdeldayem/the-fake-or-real-dataset>.

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V. CONCLUSION

An impressive 97.85% accuracy was achieved by the proposed deepfake audio detection architecture, which employed CNNs, RNNs, and MLPs in a stacking ensemble framework. This approach enhanced deepfake detection by effectively leveraging statistical descriptors, time-frequency representations, and sequential

dependencies. **The results**

AUTHORS' CONTRIBUTIONS: Parna Chaudhury conducted a comprehensive review of relevant manuscripts and prepared the initial draft of the paper. Shivani Narayan L organized and refined the manuscript, ensuring coherence and clarity throughout the text. Mridhula S investigated recent advancements in deepfake audio detection and identified potential future research directions. Ravi Tiwari reviewed the latest literature and provided critical revisions, contributing to the completion and finalization of the manuscript. All

authors read and approved the final version of the manuscript.

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